



TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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UNIT TEST - 3, NOVEMBER - 2024**CHEMISTRY**

[Max. Marks : 35]

Time Allowed : 1.30 Hours]

PART - I

Choose the correct answer.

5x1=5

Crystal field stabilization energy for high Spin d^5 octahedral complex is

- a) $-0.6 \Delta_o$ b) 0 c) $2(P-\Delta_o)$ d) $2(P+\Delta_o)$

----- is used as an antitumor drug in Cancer Treatment.

- a) Ca - EDTA b) Sodium Thio Sulphate
c) Nickel Chloride d) Cis - Platin

Assertion : Pure Iron when heated in dry air is converted with a layer of rust.**Reason** : Rust has the Composition Fe_3O_4

- a) If both Assertion and Reason are true and Reason is the correct explanation of Assertion
b) If both Assertion and Reason are true but Reason is not the correct explanation of Assertion
c) Assertion is true but Reason is false d) Both Assertion and Reason are false

The battery used in Pacemakers is -----

- a) Lead Storage battery b) Daniel Cell
c) Mercury Button Cell d) Leclanche Cell

Which of the following amines does not undergo Acetylation?

- a) t - butylamine b) ethylamine c) diethylamine d) triethylamine

PART - II

Answer any four questions. Q.No. 11 is compulsory.

3x2=6

What is Crystal field splitting energy?

Why Tetrahedral complexes do not exhibit Geometrical Isomerism?

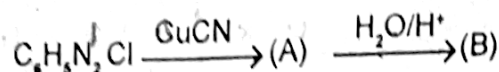
State Kohlrausch law.

What is Electro Chemical Equivalent?

Write a note on Gomberg Reaction.



11. Identify Compounds (A) and (B) in the following sequence of reactions.



PART- III

- III. Answer any three questions. Q. no. 13 is compulsory.

12. What is Linkage Isomerism? Explain with an example.
13. Write the central metal ion, Coordination number, Nature of the Ligand and Magnetic property of the complex $\text{K}_4[\text{Mn}(\text{CN})_6]$
14. State Faraday's laws of Electrolysis?
15. What are the factors affecting Electrolytic Conductance?
16. There are two Isomers with the formula CH_3NO_2 . How will you distinguish between them?
17. How will you convert Nitrobenzene into
- (i) Aniline ($1\frac{1}{2}$) (ii) Azobenzene ($1\frac{1}{2}$)

PART - IV

- IV. Answer all the questions.

18. a) Write the postulates of Werner's theory? (5)

(OR)

- b) (i) What are Ligands? Give examples? (2)
- (ii) A Solution of $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$ is green, whereas a solution of $[\text{Ni}(\text{CN})_4]^{2-}$ is colourless. Explain. (3)

19. a) Derive an expression for Nernst Equation. (5)

(OR)

- b) i) Write Debye - Huckel Onsager Equation. (2)
- ii) Write a note on Sacrificial Protection. (3)

20. a) How will you distinguish between Primary, Secondary and Tertiary aliphatic amines?

(OR)

- b) Write short notes on the following.

- A) Carbylamine Reaction ($1\frac{1}{2}$)
- B) Diazotisation ($1\frac{1}{2}$)
- C) Gabriel - Phthalimide Synthesis. (2)